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E-271 COMMUNICATIONS SYSTEM

1. SCOPE OF WORK

The work shall include the design, supply, manufacturing, factory testing, delivery to the site, installation, commissioning and testing of a complete communication system for the new steam power plant of Block 2, 3 and 4 and the associated substation.

The communication system shall be furnished complete with all specified and/or required accessories as follows:

- (1) Telephones
- (2) An adequate numbers of new or extension in solid state Private Automatic Branch Exchange (PABX). The new telephone system (new PABXs or extension whichever applicable) shall be suitable for interconnecting or integration to the existing PABX. For Block 2, shall be integrated into the existing Block 1 system, for Block 3 & 4, shall be integrated to the existing PABX of the Block 3 & 4 Gas turbine.
- (3) A public address/intercom system with solid state amplifier, speaker units, microphone stations complete with intercom channels and all accessories for steam turbine and common areas. Separate system shall be provided each for Block 2 and Block 3 & 4 system. Integration to the existing system shall be included.
- (4) A maintenance telephone system complete with telephone jacks in all station panels, all levels of the HRSG, all platform level of the stacks, in cable tunnels, the substation, and handsets. Separate system shall be provided for Block 2 and Block 3 & 4 system. A minimum 20 handsets shall be provided. Integration to the existing system shall be included.
- (5) The system shall be designed complete with all required standard interfaces for future extension of the PABXs, public address/intercom system and maintenance telephone system of the same capacity and quantity.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Section 2.5 Codes and Standards. The following specific standards are extracted from the organizations listed in Section 2.5 and are specifically applicable to this Technical Specification:

2.1 National

- PUIL : Peraturan Umum Instalasi Listrik - Indonesia
(General Regulation of Electrical Installation – Indonesia).
- SPLN : Standard Perusahaan Umum Listrik Negara.
- SII : Standard Industry Indonesia

2.2 American Society for Testing and Materials

- ASTM A123 : Zinc (Hot Galvanized) Coatings on Products
Fabricated from Rolled, Pressed and Forged
Steel Shapes, Plates, Bars and Strip
- ASTM A153 : Zinc Coating (Hot-Dip) on Iron and Steel
Hardware

2.3 International Electrotechnical Commission

- IEC 60268 : Sound system equipment
- IEC 60454 : Specifications for pressure-sensitive adhesive tapes
for electrical purposes
- IEC 60529 : Classification of degrees of protection provided by
enclosures

2.4 National Fire Protection Association

- NFPA 70 : National Electrical Code

2.5 Consulting Committee of International Telephone and Telegraph.

- CCITT G712 : Performance Characteristics of PCM Channels at
Audio Frequencies
- CCITT Q23 : Technical Features of Pushbuttons Telephone Sets
- CCITT V21 : 300 Bits per Second Duplex Modem Standardized
for Use in the General Switched Telephone Network

CCITT V22	:	1200 Bits per Second Duplexed Modems Standardized for Universal Use in the General Switched Telephone Network and on Leased Lines
CCITT V23	:	600/1200 Baud Modem Standardized for Use in the General Switched Telephone Network

3. DETAILED LIST OF EQUIPMENT TO BE PROVIDED

3.1 Telephone Instruments

The Contractor shall provide and install telephone instruments with quantity as follows as required by the Owner. The telephone instruments type shall include the following types:

- (1) Desk type *) 40 units
- (2) Indoor wall mounted type *) 25 units
- (3) Outdoor wall mounted type *) 15 units
- (4) Power bell for indoor and outdoor type *) 35 units
- (5) Acoustic both type for indoor and outdoor type *) 12 units

Note: *) The Bidder shall state these quantities in his Bid Proposal and the quantity shall be subject to approval by the Owner at the detailed design stage.

3.2 Public Address Equipment

The Contractor shall provide and install Public Address Equipment with quantity as follows as required by the Owner. The public address equipment type shall include the following:

- (1) Area PA *) 100 units
- (2) Desk microphone station type *) 20 units
- (3) Indoor wall station *) 40 units
- (4) Outdoor wall station *) 50 units

Note: *) The Bidder shall state these quantities in his Bid Proposal and the quantity shall be subject to approval by the Owner at the detailed design stage.

The number of equipment in 3.1 and 3.2 above is the total number for Block 2 system and Block 3 & 4. The number for each block shall be as directed by the Owner during detail design.

4. DETAILED REQUIREMENTS

4.1 General Requirements

All equipment shall be designed for operation in a power plant located on the sea coast in a tropical climate. Refer to Part 4, Subsections 1.1, 2.14, 2.16 for meteorological, seismic, and environmental conditions of the site.

All telephone equipment shall be of a make and model approved by the Indonesian Department of Post and Telecommunications for connections to the public telephone system.

4.2 Telephone System

4.2.1 Telephone Exchange

The Contractor shall provide a solid state telephone exchange for the station. The exchange shall be fully compatible with the Indonesian public telephone system (PT. TELKOM). The Contractor shall be responsible for connecting the plant PABX system to PT. TELKOM. It shall also be connected to tie lines provided by the Owner's line carrier equipment. A minimum of 4 (four) lines shall be provided for this interconnections. The Contractor shall be responsible for all interfacing with the line carrier equipment. The design of the exchange shall be modular so that it can be easily extended by the addition of modules or shelves.

The exchange shall be software controlled. It shall be possible to set up or change the number of a given extension, its class of service, etc. from programming console or from the operator's console in a special programming mode.

The exchange shall provide various classes of service. The classes of service shall, at a minimum, be as follows:

- (1) Internal calls only.
- (2) Internal calls plus outside calls set up by the operator.
- (3) Internal and local outside calls.

(4) Unrestricted calling.

Any station connected to the PABX shall be able to communicate with any other station by dialing a three or four digit number.

With the exception of the executive break in feature described below, any two stations shall be able to communicate without being overhead by any other station. In the case of executive break in, a distinctive tone shall be generated to indicate that the conversation is being monitored by an outside party. A distinctive tone shall also be produced if an invalid or unauthorized number is dialed.

Within the restrictions imposed by the class of service, it shall be possible to place all calls without the assistance of an operator.

Bell signals shall be intermittent and controlled automatically by the exchange without the necessity of a ringing key or push button at the telephone. The called station shall be indicated by a distinct audible tone in the receiver of the calling station.

All signalling tones, such as dial tones, busy signals, ringing tones, etc. shall be identical to those used in the Indonesian public telephone system.

No more than a single pair or wires shall be required between a station and the exchange. The exchange shall be compatible with telephone instruments identical to those used in the Indonesian public telephone system.

The exchange shall be compatible with stations using either dial pulses or dual tone (DTMF) signalling as described in CCITT Q23.

The exchange shall be capable of operating with public telephone or line carrier terminal equipment requiring either dial pulses or DTMF signalling tones. The exchange shall convert dialing signals as needed to cater to the requirements of the terminal equipment.

The exchange shall have a feature that will allow the setting up of a system of abbreviated two digit numbers for commonly dialed telephone numbers. On receipt of the predefined code, the exchange shall dial the full telephone number.

It shall be possible to add up to a total of four stations to a conversation. At least one of the stations added to a conference call may be outside of the exchange.

The exchange shall have a night service feature that will reroute incoming calls to a predetermined extension or extensions at night or when the operators are not on duty.

The exchange shall have a hunting feature so that if a station is not answered in a predetermined length of time, the call will automatically be transferred to a predetermined station.

Selected extensions shall be equipped so that an incoming outside call will be signaled by a distinctive tone in the receiver if the extension is in use at the time.

The exchange shall allow a call to be transferred from one station to another by a simple means without operator intervention.

The exchange shall be able to generate a list of outside calls made and calculate the toll for them. It shall be possible to generate the listing of toll calls sorted by extensions originating the calls. The listing shall include the telephone number of the party called, the duration of the call, the cost of the call, and the total cost of all calls made from a given extension.

The exchange shall be equipped with an executive break in feature that will allow authorized stations to break into a conversation being held at another station. It shall be possible to authorize an extension to have executive break in privileges by the use of the programming console or operator console.

The exchange shall be powered by a battery and charger combination. The Contractor shall size the battery for 6 hours of normal exchange operation without normal AC power. The exchange shall also be equipped with a separate power supply for operation from a 230/400 volt supply.

All cross talk shall be suppressed to 75 dB or more below the desired signal.

Distortion within the exchange shall be sufficiently low such that a modem operating in accordance with CCITT V.22 bis at 2400 baud shall operate successfully. The exchange shall be compatible with modems using CCITT V21, V22, or V23 signalling.

The exchange shall be equipped with an operator's station. The station shall allow the operator to perform all functions of making and answering outside calls, transferring calls, etc. by the use of a system of keys or push buttons. The console shall allow the operator to monitor the progress of calls and the status of outside lines. It shall be supplied complete with a bell or buzzer for incoming calls and an operator's handset.

The exchange shall automatically isolate faulty lines or phones left off the hook without a call being dialed. Such stations shall be disconnected from the exchange, and the faulty line indicated at the operator's console. If the telephone is again placed on hook, service shall automatically be restored.

To the extent practical, the functions of the exchange shall be divided between multiple modules, so that the failure of any one module will allow the exchange to function at reduced capacity. The Contractor shall provide details of how this is accomplished with his proposal.

The exchange shall be equipped with self diagnostic features that will detect failed modules and give a warning to the operator of trouble. It is preferred that the exchange automatically isolate such faulty modules and give an indication of the location of the failed module or modules to service personnel. The Contractor shall include any external test fixture or tester for repairing failed modules. The self diagnostic features of the exchange shall be described in the proposal.

All lines entering the exchange shall be equipped with gas tubes or a similar type of surge protection. All lines leaving the area covered by the station grounding system shall be equipped with isolating transformers, or an equivalent device approved by the Owner, rated to withstand the ultimate voltage rise of the grounding system. In addition, attention shall be given to prevent damage due to induced voltages from the 500 kV transmission lines and buses when telephone lines are routed in the vicinity of the lines and buses.

4.2.2 Telephone Instruments (Stations)

All telephone instruments shall be of a type that is fully compatible with the Indonesian public telephone system and the PABX. Instruments using DTMF signalling are preferred, and are required for all non-hazardous indoor locations.

Instruments mounted outdoors shall be contained in gasketed heavy duty enclosures that will provide a degree of protection of IPW 55 when closed.

Enclosures located in hazardous areas, shall be approved for use in such areas in accordance with IEC Standards.

The telephones shall be designed for a local loop resistance and shunt conductance to ground compatible with the PABX and the Contractor's installation.

Telephones located in noisy areas shall be mounted in acoustic booths. The booths shall cover the head and chest of a person using the telephone on three sides. The exterior of the booth shall be of a durable material with a corrosion resistant finish, compatible with the environment that it is placed in. The interior walls shall be of perforated stainless sheet. Acoustic insulation shall be provided by no less than 50 mm of a non-flammable, mildew and fungus proof, non-asbestos bearing material located between the inside and outside walls. Each booth shall have a roof to protect the telephone instrument from dripping liquids. The booth shall be equipped with a fluorescent lighting fixture located in the ceiling of the booth.

Telephones shall be equipped with power bells as indicated in Clause 3. The power bell shall use a relay operated by the ringing voltage on the telephone line to operate a 220 volt, 50 hertz power horn or howler. The power horn shall use an armature coupled to a diaphragm. The horn shall be provided with a single or double bell projector as appropriate. The power horn mechanism shall be mounted a gasketed housing with a degree of protection of IPW55. The power horn shall have an maximum output of not less than 105 dB at 3 meters. The power horn shall be equipped with a mechanism to adjust the sound output where maximum output of the horn is too loud. A minimum of 10 dB difference above the ambient noise level of the installation shall be achieved by the horn.

4.3 Public Address System

The Contractor shall provide a complete solid state public address/intercom system for the station and substation. The system shall have frequency response of 250 to 4000 hertz, flat within 3 dB, and with not more than 2.5% distortion at rated output power. The system shall be of the six channel type with a “page” channel for calling individuals, for issuing plant wide instructions, etc. by the use of loudspeakers. The other five channels shall be “party” channels for two way and group communications using handset stations. Selector switches shall provide for a choice between the “page” and “party” channels.

The public address system shall be equipped with handset stations in the - locations to be approved by the Owner at the detailed design stage. The handset stations shall consist of a hand-set and an amplifier cabinet. The handset shall be of the telephone type with a receiver and a low impedance, noise canceling transmitter designed for close talking. The handset shall be equipped with a flexible coiled cord 4 meters long. The amplifier cabinet shall contain the hang up switch, the intercom channel selector switch, and a solid state amplifier. The cabinet, when for outdoor use, shall enclose the handset behind a waterproof door and be of waterproof gasketed construction with a degree of protection of IPW55. All handset stations, except for desk mounted stations, shall be for surface mounting on walls or columns. The hang up switch shall mute nearby speakers as necessary to prevent acoustical feedback when the handset is removed from it with the selector switch in the “page” position. When the selector switch is in a “party” position, “page” calls shall come through the loud speakers.

The loudspeakers, except for offices and control rooms shall be mounted on or near their amplifier units. The amplifier cabinets shall contain a volume control and a matching transformer. The speakers and amplifiers shall be of sufficient size to amply cover the required areas at their location taking the ambient noise level into account. A minimum of 10 dB difference above the ambient noise level of the installation shall be achieved by the loudspeakers.

All amplifiers shall be equipped with an automatic gain control circuit to compensate for differences in speech. Systems using amplifiers located near their associated loudspeakers are preferred to system using a small number of large central amplifiers.

Except for speakers in office and similar area, all speakers shall be of the heavy duty industrial type with permanent magnet drivers. The speakers shall be provided with brackets for directional adjustment.

Speakers in office and administration areas shall be ceiling mounted. The speakers in control rooms shall be concealed above the louvered ceiling in such a manner that they do not cast objectionable shadows.

The Contractor shall provide all necessary line balancing assemblies required in accordance with the manufacturer's recommendations.

Public address system equipment located in the power plant and substation areas shall be operated from the power plant uninterruptible power supply.

4.4 Maintenance Telephone System

The Contractor shall provide maintenance telephone jacks as indicated in Clause 1. All telephone jacks shall be provided with spring loaded covers to prevent the entrance of dust. Additional protection shall be provided for jacks in outdoor or hazardous locations to achieve a degree of protection of IPW55.

The handsets for the maintenance telephone systems shall be self contained with 2 meter long cords. The handset shall be powered by rechargeable batteries. Battery chargers shall be provided for each battery powered handset. If the system uses a central power supply, redundant power supplies shall be provided for each circuit. The power supplies shall be protected against short circuits. They shall be operated from the station essential bus.

If the maintenance telephone system is not equipped for full duplex operation, each handset shall be equipped with a push to talk switch. There shall also be a signalling device on each handset.

The maintenance telephone handsets shall have a range of 15 km when using a normal speaking voice. It shall be possible to connect up to 25 handsets to the system at once.

The maintenance telephone system shall be classified as intrinsically safe for use in hazardous areas.

4.5 Installation

All installation work shall be performed in a manner consistent with recognized good practice for power station service and as specified herein.

There shall be no deviation or substitutions from the provisions in this specification and those of Part 4, Subsection 4.4 without the Owner's approval before such deviations or substitutions are made.

Grounding of the communication system shall be in accordance with Technical Specifications E-240 and E-241.

All floor mounted communications cabinets shall be mounted on tubular steel bases.

Wall mounted communications system equipment shall be fastened to concrete walls by means of wedge anchors and to steel columns by welding.

The raceways for the PABX, public address system, and maintenance telephone system shall be separate from those used for other purposes. Under no circumstances shall communications systems cables be installed in the same cable trays or conduits with power or control cables. Communication cables where routed through the underground site electrical duct runs shall be run in steel conduits. Separate pull pits or hand holes shall be used for communication cables.

Telephone terminal boxes shall be located in accessible locations, with the tops of the boxes 1.5 meters above the floor.

Perumtel cables shall be separated from other cables in cable trenches by a minimum of 300 mm of earth or, by 100 mm of concrete.

Communications cables buried in cable trenches that cross under roads or heavy traffic areas shall be installed in 80 mm rigid steel conduit at the crossing.

Conduits and Fittings

All conduits used inside of the power stations and in dry isolated buildings may be electric metallic tubing (EMT) for all sizes 50 mm and smaller and shall be rigid steel conduit for all sizes larger than 50 mm. Galvanized rigid steel conduits, only, shall be used for exposed installations in hazardous locations. in

wet buildings, in all outdoor areas where the conduits will be exposed to the weather, and for embedded work.

All conduits and conduit installations shall be in accordance with Technical Specification E-232.

For lighting the telephone booth, a 13 mm conduit with two 4mm² stranded copper wires shall be installed from the nearest source of 220 V lighting power.

Mounting Heights

The mounting heights of outlet boxes from the finished floor to the center line of the boxes for the communication equipment shall be as follows:

- | | | |
|-----|--|----------|
| (1) | Wall telephone, microphones, handsets and maintenance telephone jack boxes | 1,400 mm |
| (2) | Desk telephones | 450 mm |
| (3) | Public address speakers | 2,750 mm |
| (4) | Telephone power bells and horns | 2,750 mm |

The mounting heights noted above are approximate only for both concealed and exposed outlet boxes. To minimize cutting tile or brickwork, boxes concealed in tile or brick walls shall be carefully placed so that the boxes are level with the top of the tile or brick course at the approximate height noted above.

Wiring

All communications cables shall be in accordance with regulation by Indonesian Department of Post and Telecommunications for connection to the public telephone system.

- (1) Detail requirement of communication cable such as insulation, construction sizing, etc shall be in accordance with Standard Industry Indonesia for communication cable, installed above and underground.
- (2) Other requirement for cable shall be in accordance with Technical Specification E-230.

The shielding on a microphone cable must be continuous from end to end. Therefore, when splicing shielded microphone cable in junction boxes, a

minimum amount of unshielded conductor shall be exposed. After the splice is made, the unshielded area and the splice shall be wrapped with shielding tape, and the wrapping shall be soldered to the ends of the factory applied shielding on each side of the splice, to insure a continuously shielded cable.

Shielded microphone cable shall be grounded at one end only, usually at the amplifier cubicle. The manufacturer's recommendations shall be strictly followed.

Speakers shall be mounted and connected temporarily. After an audibility test, the Contractor shall mount and connect the speakers permanently.

All equipment installed outdoors shall be of the weatherproof type.

All equipment shall be cleaned and dusted regularly during construction.

4.6 Test Equipment

The Contractor shall provide all test equipment except for small hand tools necessary to service the communications system. This shall include the following:

- (1) A transistorized or vacuum tube voltmeter with an input impedance of 11 mega-ohms or greater.
- (2) A passive volt-ohmmeter with a minimum sensitivity of 20 000 ohms per volt.
- (3) A dual trace oscilloscope with a minimum vertical bandwidth of 50 MHz and triggered sweep.
- (4) A frequency counter capable of reading to 500 MHz to within 10 Hz. The counter shall be suitable for use from 50 Hz to 50 MHz without prescaling.
- (5) Bench power supplies capable of providing power for all equipment without built-in 230 volt 50 hertz supplies.
- (6) An audio frequency signal generator capable of operating from 20 hertz to at least 20 kilohertz.

- (7) Manufacturer's standard test fixture or test set for troubleshooting modules or printed circuit boards in the telephone exchange.

5. PERFORMANCE TESTING AND GUARANTEES

Type tests shall be performed on all equipment as required by the applicable industry standards and Government of Indonesia regulations.

The Contractor shall check all telephones for proper operation after installation. Also, all telephone exchange features shall be checked for proper operation.

The Contractor shall check each microphone station and loudspeaker for proper operation. Final level adjustments shall be performed with the station in normal operation and with an ambient noise level typical of normal operation.

6. QUALITY ASSURANCE/QUALITY CONTROL

Quality assurance/quality control for all communications system equipment shall conform to the requirements included in Part4, section 2, Subsection 2.7. The procedure shall include, but not be limited to, the following:

- (1) Environmental testing.
- (2) Verification of mean time between failure data.
- (3) Resistance of the exchange equipment to surges.

The quality assurance/quality control procedure specified for public address system equipment shall include, but not be limited to, environmental testing.

The documentation shall include, but not be limited to, the following:

- (1) Shop test reports.
- (2) All other documentation required by the applicable codes and standards.

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